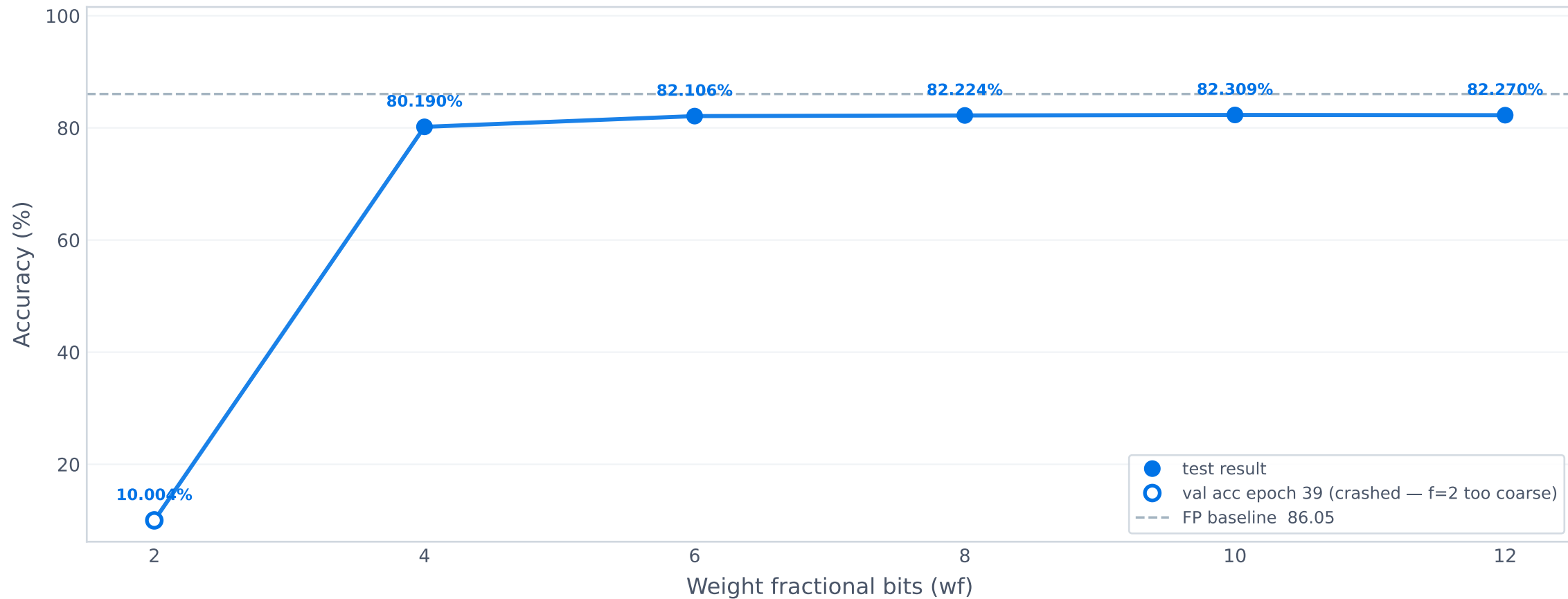


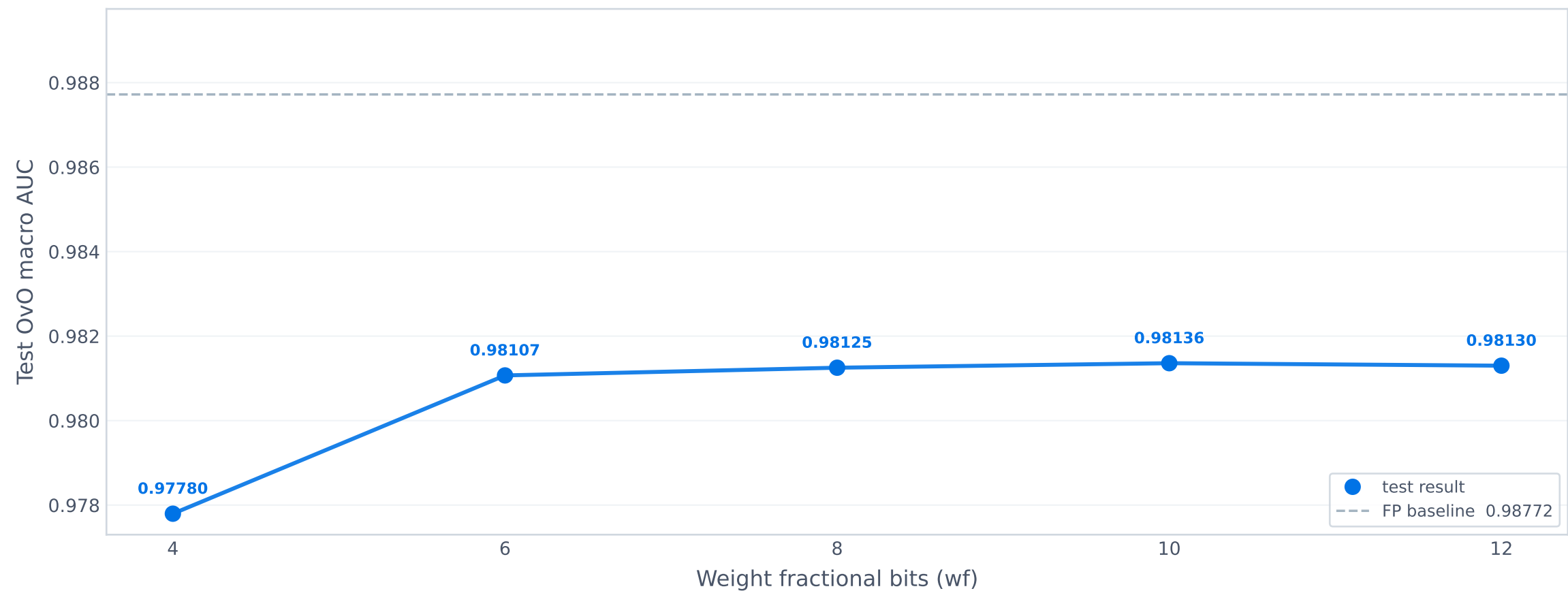
ParT weights compression — accuracy vs wf bits

Data compressed to 16 bits (f=11, i=4, k=1)



ParT weights compression — OvO AUC vs wf bits

Data compressed to 16 bits (f=11, i=4, k=1)



Overall scan metrics

Test accuracy · remaining weights · OvO macro AUC

	Accuracy (%)	Remaining weights	OvO macro AUC
Full prec.	86.051%	100.00%	0.98772
wf = 2	10.004% *	0.00%	—
wf = 4	80.190%	0.56%	0.97780
wf = 6	82.106%	1.06%	0.98107
wf = 8	82.224%	1.68%	0.98125
wf = 10	82.309%	2.07%	0.98136
wf = 12	82.270%	2.41%	0.98130

* wf=2: training ran to epoch 39 but weights collapsed to 0% by epoch 3 (2-bit quantization too coarse); eval accuracy shown is 10% (random chance for 10 classes)

Per-class AUC (OvR)

Binary one-vs-rest AUC per jet class

	QCD	Hbb	Hcc	Hgg	H4q	Hqql	Zqq	Wqq	Tbqq	Tbl
Full prec.	0.97825	0.99713	0.98790	0.98007	0.98880	0.99957	0.96695	0.98013	0.99854	0.99987
wf = 2	—	—	—	—	—	—	—	—	—	—
wf = 4	0.96754	0.99088	0.96821	0.96842	0.97866	0.99857	0.94355	0.96737	0.99533	0.99944
wf = 6	0.97099	0.99302	0.97487	0.97175	0.98196	0.99899	0.95220	0.97089	0.99644	0.99959
wf = 8	0.97113	0.99309	0.97516	0.97180	0.98185	0.99902	0.95294	0.97146	0.99647	0.99961
wf = 10	0.97125	0.99316	0.97544	0.97203	0.98210	0.99903	0.95299	0.97150	0.99650	0.99961
wf = 12	0.97121	0.99310	0.97518	0.97186	0.98200	0.99903	0.95300	0.97151	0.99651	0.99960

QCD rejection rate

1 / FPR vs QCD at per-class signal efficiency target

	Hbb @50%	Hcc @50%	Hgg @50%	H4q @50%	Hqql @99%	Zqq @50%	Wqq @50%	Tbqq @50%	Tbl @99%
Full prec.	10638.000	4149.000	123.000	1867.000	5420.000	402.000	543.000	32258.000	16260.000
wf = 2	—	—	—	—	—	—	—	—	—
wf = 4	2624.672	383.215	38.040	276.091	159.261	56.865	46.404	2339.181	88.539
wf = 6	2985.075	520.291	45.367	339.559	296.868	59.457	112.290	3012.048	163.092
wf = 8	2628.121	487.805	46.059	335.233	293.513	64.779	105.180	3372.681	158.115
wf = 10	2789.400	542.005	45.574	342.407	284.779	65.744	102.422	2894.356	151.630
wf = 12	2762.431	498.008	45.339	334.169	275.368	66.622	97.106	3215.434	159.337

Rejection = 1 / false positive rate, where FPR = fraction of QCD background jets passing the signal-score threshold at the given signal efficiency. Full prec. (FP) = full-precision uncompressed ParT model, evaluated on the 20M jet test set.